

G.C.E. (A/L) COMBINED MATHEMATICS I — PURE MATHEMATICS

UNIT: INTEGRATION (CALCULUS) — 40 STRUCTURED ESSAY QUESTIONS

TIME: 3.0 HOURS

TOTAL QUESTIONS: 40

QUESTION 01 — INTEGRATION BY PARTS & REDUCTION FORMULA

- (a) Using the method of integration by parts, evaluate the definite integral: $\int_0^{\pi/4} x \sec^2 x \, dx$
- (b) Let $I_n = \int_0^{\pi/4} \tan^n x \, dx$ for $n \geq 2$.
- (i) Prove that $I_n + I_{n-2} = \frac{1}{n-1}$
- (ii) Hence, deduce the value of I_5 .

QUESTION 02 — RATIONAL FUNCTIONS & PARTIAL FRACTIONS

Consider the integrand: $f(x) = \frac{2x^2+3x}{(x-1)(x^2+1)}$

- (a) Express $f(x)$ in partial fractions of the form: $f(x) = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$
- (b) Hence, evaluate the indefinite integral $\int f(x) \, dx$.

QUESTION 03 — TRIGONOMETRIC SUBSTITUTION & TRANSFORMATIONS

- (a) Using the substitution $x = 2 \sin \theta$, evaluate: $\int_0^{\sqrt{2}} \frac{x^2}{\sqrt{4-x^2}} \, dx$
- (b) Show that: $\int \frac{1}{1 + \sin x + \cos x} \, dx = \ln|1 + \tan(x/2)| + C$
- (Hint: Use the substitution $t = \tan(x/2)$)

QUESTION 04 — SYMMETRY PROPERTIES OF DEFINITE INTEGRALS

- (a) By using property $\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx$, prove that: $\int_0^{\pi/2} \frac{\sin \sqrt{x}}{\sin \sqrt{x} + \cos \sqrt{x}} \, dx = \frac{\pi}{4}$
- (b) Evaluate: $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$

QUESTION 05 — SUMMATION LIMITS & ALGEBRAIC INTEGRATION

- (a) Express the limit of summation as a definite integral and evaluate: $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2+r^2}$
- (b) Using substitution $u = \sqrt{x+1}$, find: $\int \frac{\sqrt{x+1}}{x} \, dx$

QUESTION 06 — DEFINITE INTEGRALS BY PROPERTIES & TRIGONOMETRIC INTEGRATION

(a) By using the property $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, evaluate: $\int_0^\pi x \sin^4 x dx$

(b) Find the indefinite integral: $\int \frac{1}{3 + 5 \cos x} dx$

(Hint: Use the substitution $t = \tan(x/2)$)

QUESTION 07 — PARTIAL FRACTIONS & INTEGRATION OF RATIONAL FUNCTIONS

Consider the algebraic fraction: $g(x) = \frac{3x^2+x+1}{(x-1)(x^2+2)}$

(a) Express $g(x)$ in partial fractions of the form: $g(x) = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$

(b) Hence, evaluate the indefinite integral $\int g(x) dx$.

QUESTION 08 — REDUCTION FORMULAE FOR DEFINITE INTEGRALS

Let $J_n = \int_0^{\pi/2} \sin^n x dx$ for $n \geq 2$.

(a) Prove the reduction formula: $J_n = \left(\frac{n-1}{n}\right) J_{n-2}$

(b) Hence, evaluate the definite integral $J_6 = \int_0^{\pi/2} \sin^6 x dx$.

QUESTION 09 — INTEGRATION BY PARTS & LOGARITHMIC FUNCTIONS

(a) Using the method of integration by parts, evaluate the definite integral: $\int_0^1 x^2 e^{2x} dx$

(b) Evaluate the indefinite integral: $\int \ln(x^2+1) dx$

QUESTION 10 — ADVANCED SUBSTITUTIONS & AREA APPLICATIONS

(a) Using the substitution $x = \sin \theta$ or otherwise, evaluate: $\int_0^1 x \sqrt{1-x^2} dx$

(b) Find the value of the indefinite integral: $\int \frac{x^3}{\sqrt{x^2+4}} dx$

QUESTION 11 — INTEGRATION BY PARTS

(a) Evaluate the definite integral: $\int_0^1 x e^{-x} dx$

(b) Using the method of integration by parts, find the indefinite integral: $\int x^2 \ln x dx$

QUESTION 12 — REDUCTION FORMULAE FOR TRIGONOMETRIC INTEGRALS

Let $I_n = \int_0^{\pi/2} \cos^n x dx$ for $n \geq 2$.

(a) Prove the reduction formula: $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$

(b) Hence, evaluate the definite integral I_5 .

QUESTION 13 — PARTIAL FRACTIONS & RATIONAL FUNCTIONS

Consider the rational function: $f(x) = \frac{4x}{(x-1)(x^2+3)}$

- (a) Express $f(x)$ in partial fractions of the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+3}$.
- (b) Hence, evaluate the indefinite integral $\int f(x) dx$.

QUESTION 14 — TRIGONOMETRIC SUBSTITUTIONS & IDENTITIES

- (a) Using the substitution $x = 3 \tan \theta$, evaluate: $\int_0^3 \frac{1}{9+x^2} dx$
- (b) Show that $\int \frac{\cos x}{1 + \sin x} dx = \ln|1 + \sin x| + C$.

QUESTION 15 — SYMMETRY PROPERTIES OF DEFINITE INTEGRALS

- (a) Using the property $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, evaluate: $\int_0^\pi x \sin x dx$
- (b) Evaluate the definite integral: $\int_{-\pi/2}^{\pi/2} x^3 \cos^3 x dx$

QUESTION 16 — LIMIT OF SUMMATION & DEFINITE INTEGRALS

- (a) Express the limit of summation as a definite integral and evaluate: $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n+r}$
- (b) Evaluate: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^2$

QUESTION 17 — ALGEBRAIC SUBSTITUTIONS

- (a) Using the substitution $u = 2x + 1$, evaluate: $\int_0^4 x \sqrt{2x+1} dx$
- (b) Find the indefinite integral: $\int \frac{x^2}{\sqrt{x^3+1}} dx$

QUESTION 18 — INTEGRATION INVOLVING INVERSE TRIGONOMETRIC FUNCTIONS

- (a) Evaluate the definite integral: $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx$
- (b) Find the indefinite integral: $\int \frac{1}{4+x^2} dx$

QUESTION 19 — FURTHER REDUCTION FORMULAE

Let $J_n = \int x^n e^x dx$ for $n \geq 1$.

- (a) Prove that $J_n = x^n e^x - n J_{n-1}$.
- (b) Hence, evaluate $\int_0^1 x^3 e^x dx$.

QUESTION 20 — INTEGRATION OF RATIONAL FUNCTIONS WITH REPEATED FACTORS

(a) Express $\frac{x}{(x-1)^2(x+2)}$ in partial fractions.

(b) Hence, find $\int \frac{x}{(x-1)^2(x+2)} dx$.

QUESTION 21 — DEFINITE INTEGRALS USING TRIGONOMETRIC SYMMETRY

(a) Prove that $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ when $f(x)$ is an even function.

(b) Using this property, evaluate $\int_{-1}^1 x^4 \sqrt{1-x^2} dx$.

QUESTION 22 — ADVANCED TRIGONOMETRIC INTEGRALS

(a) Evaluate: $\int_0^{\pi/4} \sec^4 x dx$

(b) Find the indefinite integral: $\int \tan^3 x \sec x dx$

QUESTION 23 — SUBSTITUTION METHOD WITH LOGARITHMS

(a) Evaluate: $\int_1^e \frac{\ln x}{x} dx$

(b) Find: $\int \frac{\sec^2 x}{1 + \tan x} dx$

QUESTION 24 — IMPROPER INTEGRALS & LIMITS

(a) Evaluate the definite integral: $\int_0^1 \frac{1}{\sqrt{x}} dx$

(b) Evaluate: $\int_0^\infty x e^{-2x} dx$

QUESTION 25 — PROPERTIES OF DEFINITE INTEGRALS

(a) Using $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$, evaluate $\int_0^3 |x-1| dx$.

(b) Evaluate: $\int_{-2}^2 |x^2-1| dx$

QUESTION 26 — INTEGRATION BY PARTS (REPEATED APPLICATION)

(a) Evaluate: $\int_0^\pi e^x \sin x dx$

(b) Find the indefinite integral: $\int e^{2x} \cos 3x dx$

QUESTION 27 — PARTIAL FRACTIONS WITH QUADRATIC FACTORS

Consider $h(x) = \frac{x^2+1}{(x+2)(x^2+x+1)}$.

(a) Express $h(x)$ in partial fractions.

(b) Hence, evaluate $\int h(x) dx$.

QUESTION 28 — WEIERSTRASS SUBSTITUTION ($T = \tan(X/2)$)

- (a) Using the substitution $t = \tan(x/2)$, show that $\int \frac{1}{\cos x} dx = \ln \left| \frac{1+\tan(x/2)}{1-\tan(x/2)} \right| + C$.
- (b) Evaluate: $\int_0^{\pi/2} \frac{1}{1+\cos x} dx$

QUESTION 29 — DEFINITE INTEGRALS WITH LIMITS OF SUMMATION

- (a) Evaluate using limit of a sum: $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2+4r^2}$
- (b) Evaluate: $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin\left(\frac{r\pi}{n}\right)$

QUESTION 30 — INTEGRATION OF IRRATIONAL FUNCTIONS

- (a) Using the substitution $x = \sin^2 \theta$, evaluate $\int_0^1 \sqrt{\frac{x}{1-x}} dx$.
- (b) Find $\int \frac{1}{x \sqrt{x^2-1}} dx$.

QUESTION 31 — AREA UNDER CURVES

- (a) Find the area of the region bounded by the curve $y = x^2 - 4$, the x-axis, and the lines $x = 0$ and $x = 3$.
- (b) Find the area enclosed between the curves $y = x^2$ and $y = 2x - x^2$.

QUESTION 32 — VOLUME OF REVOLUTION

- (a) Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and the line $x = 4$ is rotated through 2π radians about the x-axis.
- (b) Find the volume generated when the region bounded by $y = e^x$, $y = 0$, $x = 0$, and $x = 1$ is rotated about the x-axis.

QUESTION 33 — MISCELLANEOUS DEFINITE INTEGRALS

- (a) Evaluate: $\int_0^1 \frac{x}{1+x^4} dx$ using substitution $u = x^2$.
- (b) Evaluate: $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ using standard properties.

QUESTION 34 — RECURRENCE RELATIONS WITH DEFINITE INTEGRALS

Let $K_n = \int_0^1 x^n \sqrt{1-x} dx$ for $n \geq 1$.

- (a) Prove that $K_n = \frac{2n}{2n+3} K_{n-1}$.
- (b) Hence, find K_2 .

QUESTION 35 — ADVANCED PARTIAL FRACTIONS

(a) Express $\frac{1}{x(x^4-1)}$ in partial fractions.

(b) Hence, find $\int \frac{1}{x(x^4-1)} dx$.

QUESTION 36 — INTEGRATION INVOLVING HYPERBOLIC FUNCTIONS

(a) Evaluate: $\int_0^{\ln 2} \sinh x \, dx$

(b) Find the indefinite integral: $\int \cosh^2 x \, dx$

QUESTION 37 — INTEGRALS LEADING TO INVERSE TRIGONOMETRIC FORMS

(a) Evaluate: $\int_0^1 \frac{1}{1+x^2} dx$

(b) Find: $\int \frac{x}{\sqrt{1-x^4}} dx$

QUESTION 38 — PROPERTIES OF DEFINITE INTEGRALS (ADVANCED)

(a) Prove that $\int_0^{2a} f(x) \, dx = \int_0^a f(x) \, dx + \int_0^a f(2a-x) \, dx$.

(b) Using the above or symmetry, evaluate $\int_0^\pi x \cos^2 x \, dx$.

QUESTION 39 — INTEGRATION USING TRIGONOMETRIC PRODUCT-TO-SUM FORMULAE

(a) Evaluate: $\int_0^{\pi/2} \sin 3x \cos 2x \, dx$

(b) Find: $\int \cos 5x \cos 3x \, dx$

QUESTION 40 — COMPREHENSIVE CALCULUS REVIEW

(a) Using integration by parts, evaluate $\int_0^e (\ln x)^2 \, dx$.

(b) Find the value of $\int_0^1 \frac{x^2+2}{x+1} \, dx$ by algebraic division.